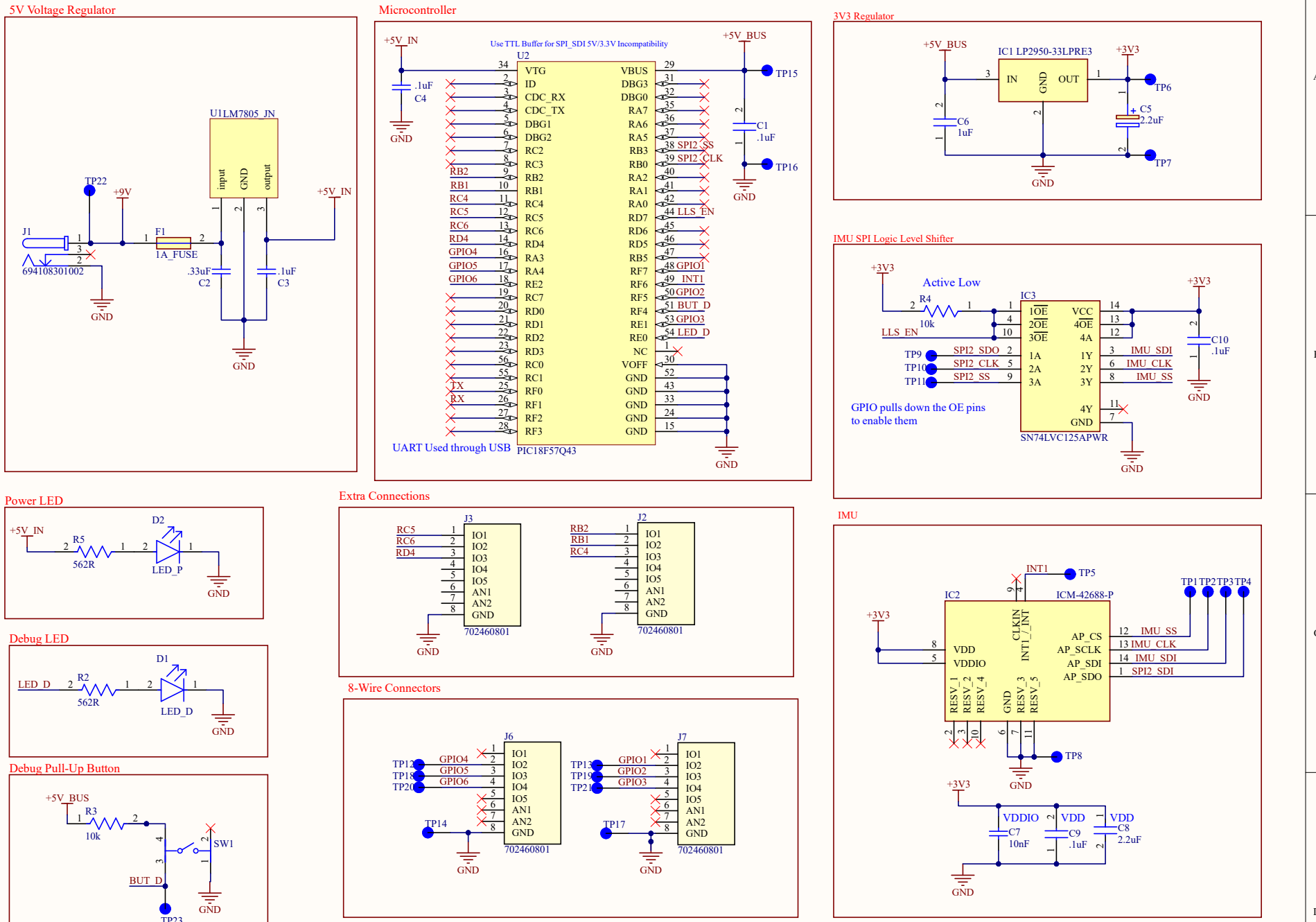

5V Voltage Regulator
U1LM7805_JN
input
GND
output
TP22
+9V
+5V_IN
J1
F1
1A_FUSE
694108301002
.33uF
C2
.1uF
C3
GND
Microcontroller
+5V_IN
+5V_BUS
Use TTL Buffer for SPI_SDI 5V/3.3V Incompatibility
U2
.1uF
C4
C1
.1uF
TP15
TP16
VTG
ID
CDC_RX
CDC_TX
DBG1
DBG2
RC2
RC3
RB2
RB1
RC4
RC5
RC6
RD4
GPIO4
GPIO5
GPIO6
RC7
RD0
RD1
RD2
RD3
RC0
RC1
TX
RF0
RX
RF1
RF2
RF3
VBUS
DBG3
DBG0
RA7
RA6
RA5
RB3
RB0
RA2
RA1
RA0
RD7
RD6
RD5
RB5
RF7
RF6
RF5
RF4
RE1
RE0
NC
VOFF
GND
SPI2_SS
SPI2_CLK
LLS_EN
GPIO1
INT1
GPIO2
BUT_D
GPIO3
LED_D
UART Used through USB
PIC18F57Q43
3V3 Regulator
+5V_BUS
IC1 LP2950-33LPRE3
IN
GND
OUT
+3V3
TP6
TP7
C6
1uF
C5
2.2uF
IMU SPI Logic Level Shifter
+3V3
Active Low
R4
10k
LLS_EN
IC3
1OE
2OE
3OE
1A
2A
3A
VCC
4OE
4A
1Y
2Y
3Y
4Y
TP9
TP10
TP11
SPI2_SDO
SPI2_CLK
SPI2_SS
IMU_SDI
IMU_CLK
IMU_SS
C10
.1uF
GPIO pulls down the OE pins
to enable them
SN74LVC125APWR
Power LED
D2
R5
562R
LED_P
Debug LED
D1
R2
562R
LED_D
Debug Pull-Up Button
R3
10k
SW1
BUT_D
TP23
Extra Connections
J3
J2
IO1
IO2
IO3
IO4
IO5
AN1
AN2
GND
702460801
8-Wire Connectors
J6
J7
TP12
TP18
TP20
TP14
TP13
TP19
TP21
TP17
IMU
IC2
ICM-42688-P
INT1
TP5
TP1
TP2
TP3
TP4
VDD
VDDIO
CLKIN
INT1 / INT
AP_CS
AP_SCLK
AP_SDI
AP_SDO
IMU_SS
IMU_CLK
IMU_SDI
SPI2_SDI
RESV_1
RESV_2
RESV_4
RESV_3
RESV_5
TP8
C7
10nF
C9
.1uF
C8
2.2uF
Title
Individual Schematic Design
Size
A
Number
Revision
Date:
10/19/2026
Sheet of
File:
C:\Users\..\egr304_board.SchDoc
Drawn By:
Joseph Nunez